



Practice Set – Discrete Distributions

Uniform, Bernoulli, Binomial, Geometric,
Negative Binomial, Poisson, Hypergeometric

STA201 – Theory of Probability and Statistics

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Instructions

- Attempt all questions. Show all necessary steps.
 - Round numerical answers to four decimal places unless stated otherwise.
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Part A:

1. A random variable X is uniformly distributed on the integers $\{2, 3, \dots, 15\}$. Find $E[X]$, $\text{Var}(X)$, and $P(X \geq 10)$.
2. A batch contains 12 components labelled $1, 2, \dots, 12$. One component is selected at random and its label X is recorded.
 - (a) Find the probability that the label is even or prime.
 - (b) Find $E[X]$ and $\text{Var}(X)$.
3. A radar system detects an incoming missile with probability 0.92. Let $X = 1$ if the missile is detected and $X = 0$ otherwise. Find $E[X]$, $\text{Var}(X)$, and the probability of a detection failure.
4. A single bit is transmitted through a noisy channel. The probability that the bit is received in error is 0.001. Let $X = 1$ denote an error.
 - (a) Write the PMF of X .
 - (b) Find $E[X]$ and $\text{Var}(X)$.
5. A machine produces items of which 12% are defective. A random sample of 25 items is drawn. Let X be the number of defectives. Find:
 - (a) $P(X = 3)$
 - (b) $P(X \leq 2)$
6. A multiple-choice examination consists of 20 questions, each with 5 possible answers of which exactly one is correct. A student who has not studied answers every question by pure guessing. Find the probability that the student gets **at least 8** correct answers.
7. In semiconductor manufacturing, 2% of the chips produced are defective. A quality engineer tests 50 chips. Find the probability that the number of defective chips is between 1 and 3 (inclusive).
8. An aircraft has three independent navigation computers. Each computer functions correctly with probability 0.9. The aircraft can navigate safely if **at least two** computers are working. Find the probability that the aircraft can navigate safely.
9. A production line produces items with a 5% defective rate. Items are inspected one by one until the first defective is found. Let X be the number of inspections required.

- (a) Find $P(X = 8)$.
 - (b) Find the expected number of inspections.
10. A message is sent over a noisy channel. The probability that a single transmission is successful is 0.7. Transmissions are independent.
- (a) Find the probability that the first successful transmission occurs on the third attempt.
 - (b) Find the probability that at most four attempts are needed for the first success.
11. A binary communication channel has a bit-error probability of 0.01. Bits are transmitted until the first error occurs. Find the probability that **more than 100 bits** are transmitted before the first error occurs. (Use the memoryless property.)
12. An oil exploration company drills wells, each of which is successful with probability 0.15, independently of the others. Let X be the k^{th} number of well on which the **fourth success** occurs.
- (a) Find $P(X = 10)$.
 - (b) Find the expected number of wells needed for 4 successes.
13. A machine has an 8% defective rate. Items are inspected until the **fifth defective** is found. Find the probability that the fifth defective occurs on the 40th item.
14. Telephone calls arrive at a customer-service centre at an average rate of 6 per minute. Let X be the number of calls arriving in a given minute.
- (a) Find $P(X = 4)$.
 - (b) Find $P(X \geq 2)$.
15. Defects on a roll of cloth occur at an average of 0.5 per square metre. Find the probability that a 3 m^2 sample contains exactly 2 defects.
16. A radioactive source emits particles at an average rate of 3 per second. Find:
- (a) the probability of no emission in a 1-second interval,
 - (b) the probability of more than 5 emissions in a 1-second interval.
17. A rare disease affects 1 person in 1000. Using the Poisson approximation to the binomial, find the probability that exactly 2 people in a random sample of 3000 have the disease.
18. A lot of 40 items contains 8 defective items. A sample of 5 items is drawn **without replacement**. Find the probability that exactly 2 are defective.
19. A committee of 4 people is to be selected at random from a group of 7 engineers and 5 managers. Find the probability that the committee contains exactly 2 engineers.

20. From a consignment of 15 transistors, 4 are faulty. Three transistors are drawn at random without replacement. Find the probability that **at most one** is faulty.

Part B: Miscellaneous – Identify the Distribution and Solve

For each of the following, identify the appropriate probability distribution (with its parameters), then answer the question.

21. A quality inspector randomly selects 30 items from a very large batch in which 4% of items are defective. What is the probability that exactly 2 are defective?
22. A batch of 15 items contains 4 defective items. Five items are drawn at random without replacement. What is the probability that exactly 2 are defective?
23. A server receives requests at an average rate of 10 per second. What is the probability that exactly 12 requests arrive in a given second?
24. A biased coin, with $P(H) = 0.4$, is tossed repeatedly until the third tail appears. What is the probability that the third tail occurs on the eighth toss?
25. Components from a production line are tested until the first failure. The failure probability of each component is 0.03, independent of the others. What is the probability that the first failure occurs on the fifteenth component?

Final Answers

Part A – Discrete Uniform

Q1. $E[X] = 8.5$, $\text{Var}(X) = 16.25$, $P(X \geq 10) = 6/14 = 3/7 \approx 0.4286$.

Q2. Even or prime = $\{2, 3, 4, 5, 6, 7, 8, 10, 11, 12\}$, so $P = 10/12 = 5/6 \approx 0.8333$.
 $E[X] = 6.5$, $\text{Var}(X) = 143/12 \approx 11.9167$.

Part B – Bernoulli

Q3. $E[X] = 0.92$, $\text{Var}(X) = 0.92 \times 0.08 = 0.0736$, $P(\text{failure}) = 0.08$.

Q4. (a) $p(1) = 0.001$, $p(0) = 0.999$. (b) $E[X] = 0.001$, $\text{Var}(X) = 0.000999$.

Part C – Binomial

Q5. (a) $P(X = 3) \approx 0.2387$. (b) $P(X \leq 2) \approx 0.4088$.

Q6. $X \sim B(20, 0.2)$. $P(X \geq 8) \approx 0.0321$.

Q7. $X \sim B(50, 0.02)$. $P(1 \leq X \leq 3) \approx 0.6131$.
(Poisson approximation with $\lambda = 1$ gives ≈ 0.6131 .)

Q8. $X \sim B(3, 0.9)$. $P(X \geq 2) = 0.243 + 0.729 = 0.972$.

Part D – Geometric

Q9. (a) $P(X = 8) = (0.95)^7(0.05) \approx 0.0349$. (b) $E[X] = 20$.

Q10. (a) $P(X = 3) = (0.3)^2(0.7) = 0.063$. (b) $P(X \leq 4) \approx 0.9919$.

Q11. $P(X > 100) = (0.99)^{100} \approx 0.3660$.

Part E – Negative Binomial

Q12. (a) $P(X = 10) = \binom{9}{3}(0.15)^4(0.85)^6 \approx 0.0160$. (b) $E[X] = 4/0.15 \approx 26.6667$.

Q13. $P(X = 40) = \binom{39}{4}(0.08)^5(0.92)^{35} \approx 0.0146$.

Part F – Poisson

Q14. (a) $P(X = 4) \approx 0.1339$. (b) $P(X \geq 2) = 1 - 7e^{-6} \approx 0.9826$.

Q15. $\lambda t = 0.5 \times 3 = 1.5$, $P(X = 2) = \frac{e^{-1.5}(1.5)^2}{2} \approx 0.2510$.

Q16. (a) $P(X = 0) = e^{-3} \approx 0.0498$. (b) $P(X > 5) \approx 0.0839$.

Q17. $\lambda = np = 3000 \times 0.001 = 3$, $P(X = 2) \approx 0.2240$.

Part G – Hypergeometric

Q18. $P(X = 2) = \frac{\binom{8}{2}\binom{32}{3}}{\binom{40}{5}} \approx 0.2111$.

Q19. $P(X = 2) = \frac{\binom{7}{2}\binom{5}{2}}{\binom{12}{4}} = \frac{210}{495} \approx 0.4242$.

Q20. $P(X \leq 1) = \frac{\binom{11}{3} + \binom{4}{1}\binom{11}{2}}{\binom{15}{3}} = \frac{165 + 220}{455} \approx 0.8462$.

Part H – Miscellaneous

Q21. Distribution: Binomial($n = 30, p = 0.04$). $P(X = 2) \approx 0.2220$.

Q22. Distribution: Hypergeometric($N = 15, K = 4, n = 5$). $P(X = 2) = \frac{\binom{4}{2}\binom{11}{3}}{\binom{15}{5}} \approx 0.3297$.

Q23. Distribution: Poisson($\lambda = 10$). $P(X = 12) \approx 0.0948$.

Q24. Distribution: Negative Binomial($r = 3, p = 0.6$). $P(X = 8) = \binom{7}{2}(0.6)^3(0.4)^5 \approx 0.0465$.

Q25. Distribution: Geometric($p = 0.03$). $P(X = 15) = (0.97)^{14}(0.03) \approx 0.0196$.